Case Study

Mobile Computing

Wireless Markup Language

July 06, 2026

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Teacher's Signature

WRITE A WML PROGRAM TO CREATE A CARD.

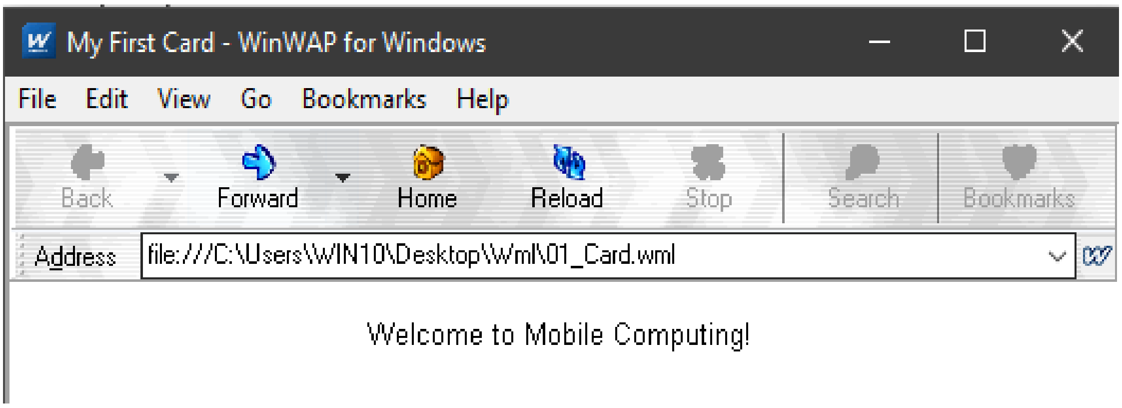
INTRO:

A WML card is the basic unit of display in a Wireless Application Protocol (WAP) browser. Each card can contain text, images, and input elements, and is defined inside a `<card>` tag within a `<deck>`. This program creates a simple card that displays a welcome message.

CODE:

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
 <card id="card1" title="My First Card">
      <p align="center">
            Welcome to Mobile Computing!
      </p>
 </card>
</wml>
```

OUTPUT



WRITE A WML PROGRAM TO CREATE A DECK THAT CONTAINS TWO CARDS AND PROVIDE THE FUNCTIONALITY OF CALLING TWO CARDS FROM ONE ANOTHER.

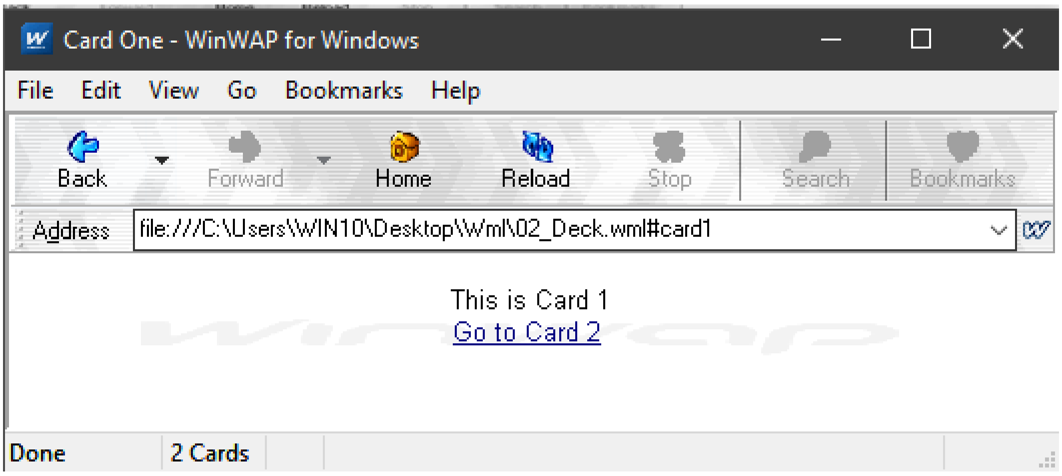
INTRO:

A WML deck can contain multiple cards, and navigation between them is done using `<go>` and `<prev>` tags. This program creates two cards with links to jump back and forth between them, demonstrating basic inter-card navigation.

CODE:

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
      <card id="card1" title="Card One">
            <p align="center">
                  This is Card 1<br/>
                  <a href="#card2">Go to Card 2</a>
            </p>
      </card>
      <card id="card2" title="Card Two">
            <p align="center">
                  This is Card 2<br/>
                  <a href="#card1">Go to Card 1</a>
            </p>
      </card>
</wml>
```

OUTPUT



WRITE A WML PROGRAM TO IMPLEMENT THE FUNCTIONALITY OF LOGIN BY USERNAME.

INTRO:

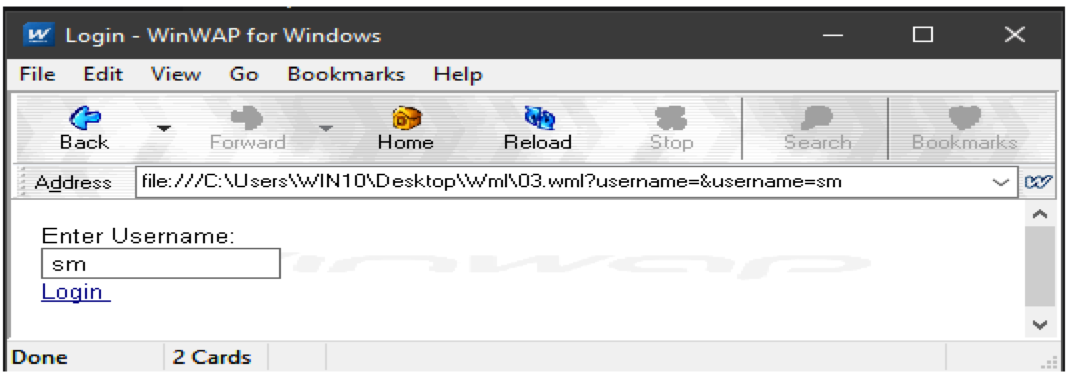
Wml supports user input through `<input>` fields. This program takes a username from the user and displays a welcome message on the next card using a `<go>` tag with sendreferer or bypassing variables via postfield. Here we use a simple approach with # navigation and variable passing.

CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.3//EN"
"http://www.wapforum.org/DTD/wml13.dtd">
<wml>
  <card id="card1" title="Practical_8">
    <p>Welcome to registration page......</p>

    <do type="unknown" label="LOGIN">
      <go href="#card2" />
    </do>
  </card>
  <card id="card2" title="TIMER">
    <p align="center">
      Username: <input type="text" name="username" /> Password:
      <input type="password" name="password" />
    </p>
    <do type="unknown" label="login">
      <go href="#card3" />
    </do>
  </card>
  <card id="card3" title="Data">
    <p>Your UserName is: $username</p>
    <p>Your Password is: $password</p>
  </card>
</wml>
```

OUTPUT



WRITE A WML PROGRAM TO DISPLAY SPECIAL CHARACTERS ON THE SCREEN.

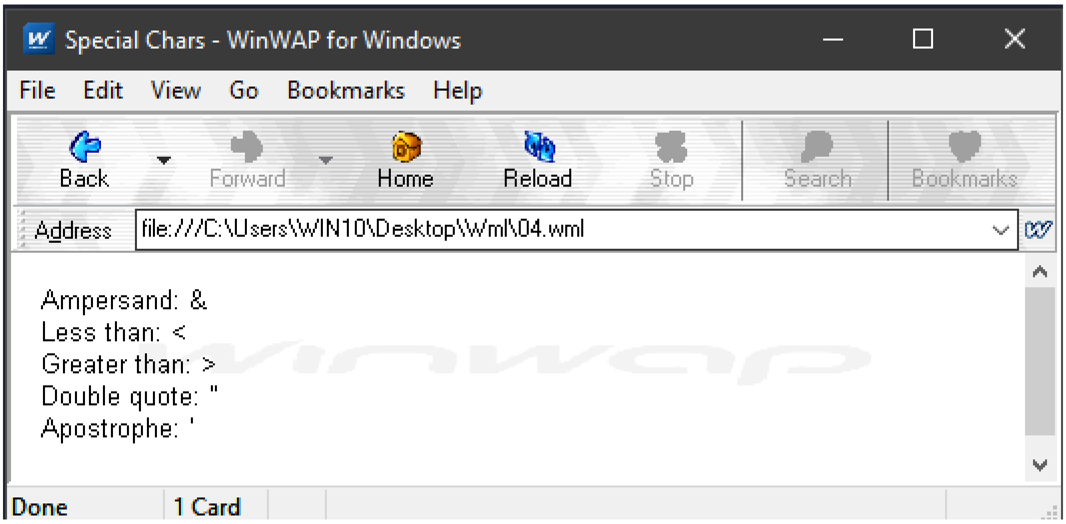
INTRO:

Special characters like &, <, >, ", and ' need to be escaped in WML using entities (e.g., &, <, >, ", '). This program demonstrates how to display them properly.

CODE:

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="special" title="Special Chars">
    <p>
      Ampersand: &amp;<br/>
      Less than: &lt;<br/>
      Greater than: &gt;<br/>
      Double quote: &quot;<br/>
      Apostrophe: &apos;
    </p>
  </card>
</wml>
```

OUTPUT:



WRITE A WML PROGRAM TO DISPLAY AN IMAGE ON THE SCREEN AFTER 5 SECONDS.

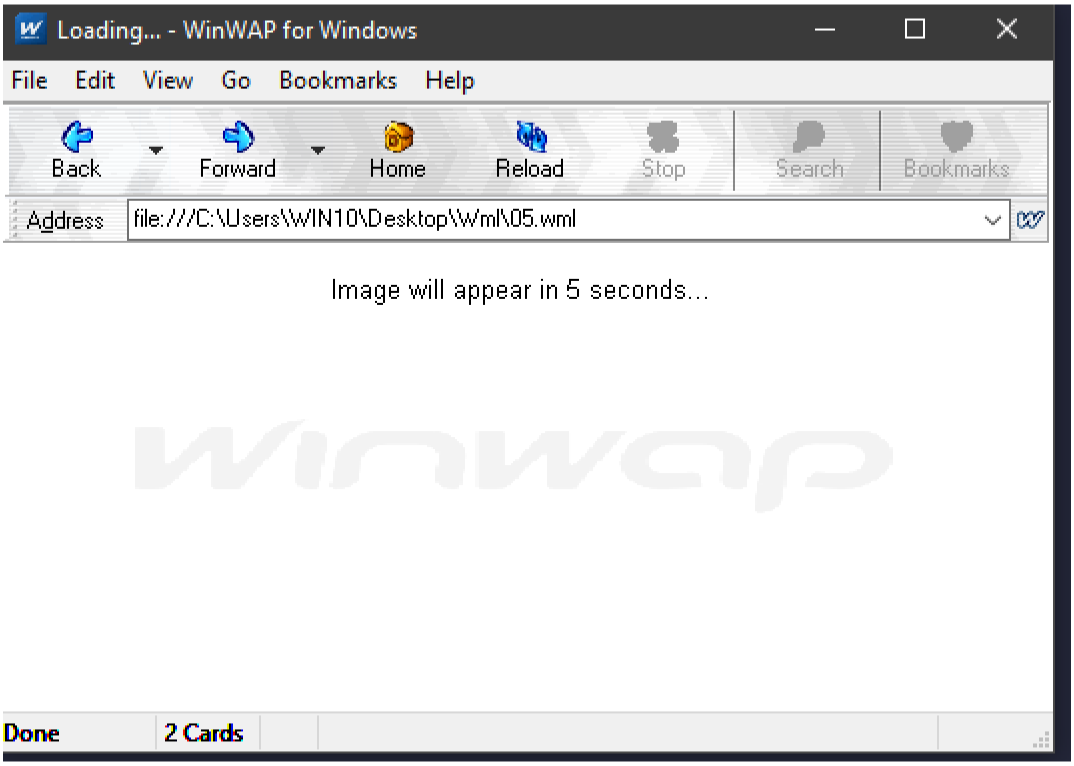
INTRO:

WML supports timers using the `<timer>` tag inside a card. The ontimer event triggers navigation to another card after a specified time (in tenths of seconds). This program shows a text message for 5 seconds, then navigates to a card that displays an image.

CODE:

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="wait" title="Loading..." ontimer="#imagecard">
    <timer value="50"/>
    <p align="center">
      Image will appear in 5 seconds...
    </p>
  </card>
  <card id="imagecard" title="Image Display">
    <p align="center">
      <img src="sample.wbmp" alt="Sample Image"/>
    </p>
  </card>
</wml>
```

OUTPUT:



WRITE A WML PROGRAM TO DEVELOP A CALCULATOR.

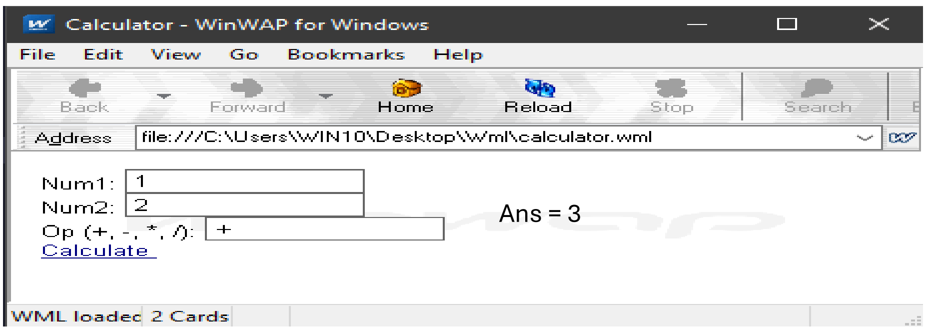
INTRO:

WML can perform simple arithmetic using the calc library. This program takes two numbers and an operator from the user and computes the result using WMLScript's eval or Lang.float() functions.

CODE:

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="calc" title="Calculator">
    <p>
      Num1: <input name="n1" type="text" format="*N"/><br/>
      Num2: <input name="n2" type="text" format="*N"/><br/>
      Op (+, -, *, /): <input name="op" type="text" format="*N"/><br/>
      <anchor>Calculate
        <go href="#result">
          <postfield name="n1" value="$(n1)"/>
          <postfield name="n2" value="$(n2)"/>
          <postfield name="op" value="$(op)"/>
        </go>
      </anchor>
    </p>
  </card>
  <card id="result" title="Result">
    <p>
      <onevent type="onenterforward">
        <refresh>
          <setvar name="res" value="$(n1) $(op) $(n2)"/>
        </refresh>
      </onevent>
      $(n1) $(op) $(n2) = <br/>
      <!-- We use WMLScript for actual calculation -->
      <a href="calc.wmls#compute($(n1),$(n2),$(op))">Compute</a>
    </p>
  </card>
</wml>
```

OUTPUT:



WRITE A WML PROGRAM TO DEVELOP A SALARY CALCULATOR.

INTRO:

This salary calculator takes the basic salary, allowances, and deductions as input, then computes the net salary using a WMLScript function.

FILE 1: SALARY.WML

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="salary" title="Salary Calculator">
    <p>
      Basic Salary: <input name="basic" type="text" format="*N"/><br/>
      Allowances: <input name="allow" type="text" format="*N"/><br/>
      Deductions: <input name="deduct" type="text" format="*N"/><br/>
      <anchor>Calculate
        <go href="salary.wmls#compute($(basic),$(allow),$(deduct))"/>
      </anchor>
    </p>
  </card>

  <card id="result" title="Net Salary">
    <p>
      Net Salary: $(net)<br/>
      <a href="#salary">Back</a>
    </p>
  </card>
</wml>
```

FILE 2: SALARY.WMLS

```
extern function compute(basic, allow, deduct) {
    var b = Lang.parseFloat(basic);
    var a = Lang.parseFloat(allow);
    var d = Lang.parseFloat(deduct);
    var net = b + a - d;

    WMLBrowser.setVar("net", String(net));
    WMLBrowser.go("salary.wml#result");
}
```

WRITE A WML PROGRAM TO DEVELOP A LOAN CALCULATOR.

INTRO:

This loan calculator computes the monthly EMI (Equated Monthly Installment) using the formula:
EMI = P * r * (1+r)^n / ((1+r)^n - 1) where P = principal, r = monthly interest rate, n = number of months.

FILE 1: LOAN.WML

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="loan" title="Loan Calculator">
    <p>
      Principal: <input name="principal" type="text" format="*N"/><br/>
      Rate (%% p.a.): <input name="rate" type="text" format="*N"/><br/>
      Months: <input name="months" type="text" format="*N"/><br/>
      <anchor>Calculate EMI
        <go href="loan.wmls#compute($(principal),$(rate),$(months))"/>
      </anchor>
    </p>
  </card>
  <card id="result" title="Monthly EMI">
    <p>
      EMI: $(emi)<br/>
      <a href="#loan">Back</a>
    </p>
  </card>
</wml>
```

FILE 2: LOAN.WMLS

```
extern function compute(p, r, n) {
    var P = Lang.parseFloat(p);
    var R = Lang.parseFloat(r) / 1200;  // monthly interest rate
    var N = Lang.parseFloat(n);
    var emi;

    if (R == 0) {
        emi = P / N;
    } else {
        var temp = Math.pow(1 + R, N);
        emi = P * R * temp / (temp - 1);
    }

    WMLBrowser.setVar("emi", String(emi));
    WMLBrowser.go("loan.wml#result");
}
```

WAP TO DISPLAY AN IMAGE IN WML.

INTRO:

Images in WML are displayed using the <img> tag with the src attribute pointing to a .wbmp (Wireless Bitmap) file. This program shows a simple image on the screen.

CODE: IMAGE.WML

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="imagecard" title="Image Display">
    <p align="center">
      <img src="sample.wbmp" alt="Sample Image"/>
      <br/>
      This is an image
    </p>
  </card>
</wml>
```

[Note: You need a .wbmp image file in the same folder. You can convert any image to .wbmp using online converters or tools like wbmpconv.]

WRITE A WML PROGRAM TO DISPLAY THE TEXT ON THE SCREEN AFTER 5 SECONDS.

INTRO:

This program uses the <timer> tag to wait 5 seconds (50 tenths of a second) and then automatically navigates to another card that displays the desired text.

CODE: DELAYED_TEXT.WML

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN"
"http://www.wapforum.org/DTD/wml_1.1.xml">
<wml>
  <card id="wait" title="Loading..." ontimer="#display">
    <timer value="50"/>
    <p align="center">
      Please wait 5 seconds...
    </p>
  </card>
  <card id="display" title="Delayed Text">
    <p align="center">
      Hello! This text appeared after 5 seconds.
    </p>
  </card>
</wml>
```